Dhana Karthikeya Ventrapragada

Berkeley, CA

+1(669)269-8955 · vdkarthikeya@berkeley.edu · Portfolio · LinkedIn · GitHub

Education

University of California, Berkeley **Dec 2027**

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Data Structures · Structure & Interpretation of Computer Programs · Machine Learning · Data Engineering · Optimization Models · Principles & Techniques of Data Science · Probability for Data Science

Experience

Engineering Intern **May 2026 – Present**

Guang Labs *Remote*

- Built an **async video/image domain-classification API** (**FastAPI**, **Auth0**, **Postgres** job store): submit-and-poll **REST** endpoints run classification as **background jobs** with persisted status and granular per-item progress.
- Implemented the Python pipeline end-to-end: resolving media from **4 source types** (local, **S3**, HTTP), auto-detecting image vs. short/long video via **ffprobe/ffmpeg**, and classifying via **Gemini** multimodal calls with a trust-gated **resample loop** over fresh windows.
- Split the service into isolated modules (resolver, sampler, trust, Gemini/fake providers, **Pydantic** schemas) behind a central orchestrator, and drove the production path to done through code review, backed by **86 unit, integration, and acceptance tests**.

Data Science Intern **Jun 2026 – Present**

IDX Exchange *Remote*

- Built a **preprocessing pipeline** for a California home-price model on **70K+ CRMLS single-family records** — data cleaning, statistical outlier caps on lot size and price, and a **chronological train/test split** holding out the latest month.
- Diagnosed and fixed a **train/test data-leakage bug** in the outlier-threshold logic (thresholds were computed over the full dataset), recomputing them from training data only; benchmarked Linear Regression, Decision Tree, and Random Forest, with **Random Forest** reaching R^2 **0.85** and **MdAPE 8.8%** on the held-out month.

Undergraduate Course Staff (CS 61A) — UCS1 **Jun 2026 – Present**

University of California, Berkeley *Berkeley, CA*

- Course staff for **CS 61A** (**200+ students**), an intro CS course taught in **Python and Scheme** — teaching **2 weekly exam-prep sections**, authoring/deploying Gradescope assignments, and adjudicating grading, regrade, and extension cases in weekly office hours.

Projects

Movie Genre Classification (Multi-Label NLP) | *Python, PyTorch, SentenceTransformers*

- Built a **multi-label genre prediction system** on **42K+ MovieLens/TMDB films**, parsing nested IMDb metadata with regex, engineering log budget/revenue and temporal features, and embedding title+tagline+overview text via **MiniLM, MPNet, and e5-large-v2**.
- Benchmarked **Logistic Regression, XGBoost, and LightGBM** on 1024-dim e5 embeddings; combined a rare-genre cutoff with threshold tuning to address class imbalance, lifting **Macro F1 from 0.48 to 0.66** (Micro F1 0.68) with Logistic Regression beating tree models while training ∼12× faster.

Email Spam/Ham Classifier | *Python, scikit-learn, pandas, NumPy*

- Engineered **50+ features** from raw emails — regex counts (links, punctuation, dollar signs), HTML/subject indicators, log-scaled length features, and a curated spam/ham bag-of-words.
- Trained **L1-regularized Logistic Regression** tuned via **GridSearchCV** and validated with 10-fold cross-validation, achieving **92.1% test accuracy** on the held-out leaderboard set.

Housing Price Prediction & Assessment Bias Analysis | *Python, pandas, NumPy, scikit-learn*

- Built a **linear regression pipeline** on **200K+ Cook County Assessor (CCAO) records**, engineering log-transformed sqft, **polynomial age & geo terms, latitude×longitude interactions**, and one-hot road/garage features; filtered non-arm's-length $1 sales to align with real assessment practice.
- Achieved **test RMSE = 0.562** via 4-fold cross-validation, then quantified **systemic over- and under-estimation across price bands**, connecting design choices to documented racial inequities in property tax assessment.

Technical Skills

Languages: Python, Java, C++, SQL, Scheme
Backend & Infrastructure: FastAPI, Pydantic, PostgreSQL, REST APIs, Auth0, AWS S3, Gemini API, pytest, ffmpeg
ML & Data Science: pandas, NumPy, scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, SentenceTransformers
Developer Tools: Git, GitHub, VS Code, IntelliJ, Jupyter Notebook, Google Colab, MATLAB